

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (CE/IT) Examination

MAY 2014

IT313 Programming Embedded Systems

Date: 15.05.2014, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) In context of RTOS, multitasking and multiprocessing are more important. Explain [03]
multiprocessing and multitasking.
- (b) With the growth of the use of commercial off-the-shelf real-time operating systems, [04]
the terms task, thread and process are widely used. Compare thread and process.
- Q - 2 (a) What is Embedded firmware? What are the different approaches available for [07]
embedded firmware development?
- (b) What is hardware software co-design? Explain fundamental issues in hardware [07]
software co-design.

OR

[07]

- (b) Enumerate the various factors to be considered, while selecting a microcontroller for
an embedded system design.
- Q - 3 (a) What is the difference between microcontrollers and microprocessors? Explain the [07]
role of microcontrollers and microprocessors in embedded system design.
- (b) Explain the different types of preemptive scheduling algorithms. State the merits and [07]
de-merits of each.

OR

- (b) What are the different files generated during the cross compilation of an Embedded C [07]
file? Explain them in detail.

SECTION – II

- Q - 4 (a) What is an embedded system? Explain the different applications of embedded [03]
systems.

- (b) Justify. "The cost of designing an embedded system may be thousands of times the cost of its processor and hardware units." [04]

Q - 5 (a) Interface LM35 through ADC0804 with 8051. Write an 8051 C program to display temperature on Sevensegment [06]

- (b) What is EDLC? Why EDLC is essential in embedded product development? Explain the different Life Cycle Models adopted in embedded product development. [08]

OR

- (b) Write an 8051 C program to check whether password is correct or not. Set 4-digit password & confirm it. Enter password through keypad and display message on LCD. [08]

Q - 6 (a) List the different communication buses used in embedded system design. With help of neat sketch explain I2C bus and SPI bus protocol. Also differentiate I2C and SPI. [07]

- (b) Interface DS1307(RTC) with 8051. Write an 8051 C program to set date 14/02/2014. Transfer it serially with baud rate 9600 or display on LCD. XLAT = 11.0592 MHz. [07]

OR

Q - 6 (a) Write a Kernel program to display "Hello World!" [04]

- (b) Interface DS1307 (RTC) with 8051. Write an 8051 C program to set and date through keypad. Also read time and date from RTC and display on LCD as well as serial port. [10]
